MICROFI COLLECT

Test Documentation

IEEE 829–Style Verification & Validation Pack

Test Plan · Design · Cases · Procedures
Transmittal · Log · Incidents · Summary

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Design system aligned with Business Analysis v3.1

0. Contents

1. Document Map

This pack follows the classical IEEE 829 test documentation set, adapted to MICROFI Collect (MVP). Each component below is a full section of this document.

#	Component	Purpose in this pack
1	Test Plan	Approach, scope, resources, schedule, deliverables
2	Test Design Specification	Features, conditions, coverage per module
3	Test Case Specification	Step-by-step cases with expected results
4	Test Procedure Specification	Ordered execution of related case groups
5	Test Item Transmittal Report	Handoff of builds from development to test
6	Test Log	Chronological execution record
7	Test Incident Report	Deviations / defects for investigation
8	Test Summary Report	Overall assessment for stakeholders

References: Business Analysis & Requirements; System Architecture & Software Design; Offline-First Geolocation Spec; OpenAPI / Swagger.

2. Test Plan

Identifier

`TP-MICROFI-V1.0` — Test Plan for MICROFI Collect Minimum Viable Product.

Introduction

MICROFI is a hybrid offline-first platform for itinerant microfinance collection in the CEMAC zone. This plan governs verification of MVP flagships: secure agent terminal, escrow lockout, offline collection with GPS gate and denomination, digital OFJ, and client read views.

Test items

- Flutter agent application (Android)
- Flutter client application (read profile/balance)
- Next.js back office (enrollment, OFJ, ceilings)
- Core Backend (Spring Boot 4) and public API via Gateway
- CBS Middleware mock
- Docker infrastructure: PostgreSQL, Redis, RabbitMQ

Features to be tested (in scope)

FR-01, FR-02, FR-04, FR-06, FR-07, FR-08, FR-10, FR-12, FR-16, FR-17, FR-18 (stub), FR-20, FR-21; BR-01, BR-02, BR-03, BR-05; NFR-01, NFR-08, NFR-09 (pulse behaviour).

Features not to be tested (V1.0)

Deep multi-bank GIMAC automation (FR-03 full); production CBS vendor certification; national-scale load; iOS; production USSD gateway.

Approach

1. **Unit** — validators (escrow, denomination, GPS presence, JWT).
2. **Integration** — Core with Postgres/Redis/RabbitMQ; sync and middleware mock.
3. **API** — Swagger UI / OpenAPI (springdoc); positive and negative cases.
4. **System / E2E** — physical Android preferred for GPS and offline.
5. **Acceptance** — scripted demo (Section 5.8).

Item pass / fail criteria

- **Pass:** Expected result observed; no Critical defect on money path or GPS gate.
- **Fail:** Deviation from expected; logged as incident (Section 8).
- **Blocked:** Environment or dependency prevents execution; recorded in Test Log.

Suspension / resumption

Suspend formal acceptance if Critical defect on escrow, GPS gate, or double-posting remains open. Resume after fix is transmitted (Section 6) and smoke re-run.

Test deliverables

- This document (plan, design, cases, procedures, templates)
- Executed Test Log and Incident Reports
- Test Summary Report at MVP freeze
- Swagger export / OpenAPI snapshot used for the run

Environmental needs

Docker Compose stack; JDK 21; Flutter 3.x; Node 20; physical Android with GPS; optional Bluetooth printer.

Responsibilities

Role	Responsibility
Author / developer	Build features, unit/integration tests, defect fixes
Tester / author	Execute procedures, log results, file incidents
Supervisor	Accept demo; approve summary report
MFI liaison (optional)	Operational realism of OFJ and collection

Staffing and training

Solo or small academic team; training = this pack + BA use cases + Swagger walkthrough.

Schedule (indicative, aligned to 6-week MVP)

Phase	Test focus	Week
0 Foundation	Health, gateway, migrations	1
1 Auth + escrow	TC-AUTH, TC-ESC	2
2 Collection + GPS	TC-COL	3
3 Offline sync	TC-OFF	4
4 OFJ + client	TC-OFJ, TC-CLI	5
5 Harden + acceptance	TC-GEO, TC-SEC, demo procedure	6

Risks and contingencies

GPS / emulator

Use physical devices for all P0 COL/GEO cases.

CBS mock only

Vendor UAT scheduled when credentials exist; not a V1.0 exit blocker.

Performance & accessibility (future)

National-scale load and formal accessibility audits are out of scope for V1.0 formal tests. Track as V1.1+ work: API latency budgets under sync bursts; WCAG-oriented review of back office.

Approvals

Author: ________________	Supervisor: ________________	Date: ____________

3. Test Design Specification

Identifier

`TDS-MICROFI-V1.0`

Features to be tested (design view)

Testing is organised by functional module. Each feature is refined into **test conditions** (what must be true) and **coverage notes**.

AUTH — Authentication & enrollment

- **Conditions:** Valid PIN+IMEI; invalid PIN; IMEI mismatch; admin can enroll.
- **Coverage:** FR-01, FR-02, UC-01, UC-02.
- **Cases:** TC-AUTH-01..04.

ESC — Escrow & lockout

- **Conditions:** Under ceiling; at/over ceiling; offline uses last synced remaining; top-up restores; override (P1).
- **Coverage:** FR-04, BR-03, UC-04.
- **Cases:** TC-ESC-01..05.

COL — Collection, denomination, GPS gate

- **Conditions:** Happy path; GPS off; bad accuracy; denomination mismatch; client ID methods; immutable geotag.
- **Coverage:** FR-06, FR-08, FR-12, BR-02, BR-05.
- **Cases:** TC-COL-01..06.

OFF — Offline & sync

- **Conditions:** Encrypted queue; idempotent sync; multi-tx burst; failed sync retains queue.
- **Coverage:** FR-07, NFR-01, NFR-08.
- **Cases:** TC-OFF-01..04.

GEO — Continuous tracking

- **Conditions:** 5-min pulse offline; batch 202; map last-known; off-duty stop (P1).
- **Coverage:** FR-10, FR-11, NFR-09, NFR-10.
- **Cases:** TC-GEO-01..04.

OFJ — End of day

- **Conditions:** $\Delta = 0$ close; negative variance; positive per policy; export stub.
- **Coverage:** FR-16, FR-17, FR-18, BR-01.
- **Cases:** TC-OFJ-01..04.

CLI — Client

- **Conditions:** Own profile only; balance after sync; history (P1).
- **Coverage:** FR-20, FR-21, FR-22.
- **Cases:** TC-CLI-01..03.

SEC — Security & gateway

- **Conditions:** No JWT; branch isolation; rate limit; encrypted offline DB.
- **Coverage:** NFR-01, STRIDE (Elevation, Disclosure, DoS).
- **Cases:** TC-SEC-01..04.

Approach refinements

Negative testing is mandatory on money path and GPS gate. Offline scenarios use airplane mode. API exploration and contract checks use **Swagger UI / OpenAPI (springdoc)**.

Feature pass criteria

A feature passes when all linked **P0** cases Pass and no open Critical incident remains for that feature.

4. Test Case Specification

> **Convention.** `TC-MODULE-NN`. Every case below includes all five fields: **Objective**, **Preconditions**, **Steps**, **Expected**, **Refs**.

Prioritization method

Priorities are **risk-based**, aligned with the BA Feature Priority Matrix and money-path safety:

- **P0 — MVP blocker:** Failure allows incorrect money movement, double posting, lockout bypass, collection without GPS, or blocks the acceptance demo. Must Pass before freeze.
- **P1 — Fast-Follow:** Important for pilot quality (overrides, history, rate limit, map polish) but not required for V1.0 academic acceptance.
- **P2 — Backlog:** Deferred with FR backlog (full multi-bank, advanced analytics).

Coverage metrics (catalogue size)

Category	Count	Notes
Total test cases in catalogue	38	Including boundary & usability
P0	28	MVP exit set
P1	10	Fast-Follow
P2	0	None in this pack
Planned automation (V1)	11 of 28 P0 (≈39%)	See breakdown below

Automation breakdown (P0).

- **Automatable on CI (API/unit) — 11 cases:** TC-AUTH-02 (API reject); TC-ESC-01,02,06,07,08 (escrow rules with fixtures); TC-COL-04 (denomination sum); TC-OFF-02,04 (idempotency / retry with mock); TC-SEC-01,02 (JWT and BOLA via HTTP).
- **Device / manual P0 — 17 cases:** TC-AUTH-01,03,04; TC-ESC-03; TC-COL-01,02,03,05,06; TC-OFF-01,03; TC-GEO-01,02; TC-OFJ-01..04; TC-CLI-01,02; TC-SEC-04; TC-UX-01. Require physical GPS, airplane mode, encrypted file inspect, or UI timing.
- **Method:** JUnit/WebTestClient + Swagger for the API set; device checklist for the remainder before acceptance.

Seed data (shared preconditions)

> Branch `BR-YDE-01`; agents A1 (ceiling 100,000 XAF), A2 (50,000 XAF); clients C01–C10; BEAC denomination grid. No production PII.

TC-AUTH

TC-AUTH-01 P0

Agent login success
Objective: Valid agent authenticates on bound device.
Preconditions: Enrolled; IMEI matches; PIN correct.
Steps: Open app → enter PIN → submit.
Expected: JWT; home screen; secure session storage.
Refs: FR-01, UC-01.

TC-AUTH-02 P0

Wrong PIN rejected
Objective: Invalid credentials never yield a session.
Preconditions: Agent A1 enrolled with known correct PIN; device IMEI matches.
Steps: 1) Open agent app. 2) Enter an incorrect PIN. 3) Submit. 4) Repeat until configured attempt limit if applicable.
Expected: Access denied on each attempt; no JWT issued; optional lockout message after limit; correct PIN still works afterward unless permanently locked by policy.

Refs: FR-01.

TC-AUTH-03 P0

IMEI mismatch rejected

Objective: Agent cannot authenticate from an unbound device.
Preconditions: Agent A1 bound to IMEI-A in back office; test build reports IMEI-B (second device or mocked IMEI).
Steps: 1) Install/open agent app on IMEI-B device. 2) Enter A1's correct PIN. 3) Submit.
Expected: Login refused; message indicates device not authorised; no JWT; no access to collection screens.
Refs: FR-01.

TC-AUTH-04 P0

Admin enrolls agent

Objective: Back office can provision a new field agent.
Preconditions: Admin user logged into back office; branch `BR-YDE-01` exists.
Steps: 1) Create agent with name, branch, IMEI, initial PIN. 2) Set escrow ceiling. 3) Save. 4) On matching device, login with that PIN.
Expected: Agent appears in list; login succeeds on bound device; escrow value visible to agent or admin.
Refs: FR-02, UC-02.

TC-ESC

TC-ESC-01 P0

Collection allowed under ceiling

Objective: Valid deposit within remaining escrow is accepted.
Preconditions: A1 logged in; remaining escrow $\geq$15,000 XAF; client C01; GPS on.
Steps: 1) Identify C01. 2) Enter amount 5,000 XAF. 3) Enter denomination lines summing to 5,000. 4) Confirm.
Expected: Success feedback; remaining escrow decreased by 5,000; transaction visible in history.
Refs: FR-04, BR-03.

TC-ESC-02 P0

Lockout when cumulative reaches ceiling

Objective: No further collection when remaining escrow is exhausted.
Preconditions: A2 (or A1) remaining escrow $= 0$ after prior collections or fixture.
Steps: 1) Open collection screen. 2) Attempt to start or confirm a new deposit of 1,000 XAF with GPS on.
Expected: Collection control disabled or API rejects; clear lockout message; no new financial row; remaining still 0.
Refs: FR-04, BR-03, UC-04.

TC-ESC-03 P0

Offline lockout uses last synced ceiling

Objective: Device must not queue amounts above last known remaining while offline.
Preconditions: Last successful sync showed remaining $= 5{,}000$ XAF; enable airplane mode.
Steps: 1) Confirm offline indicator. 2) Attempt collection of 6,000 XAF with denomination and GPS.
Expected: Local validation rejects; transaction not stored as valid pending collection; remaining local copy unchanged.
Refs: FR-04, FR-07, BR-Lock-01.

TC-ESC-04 P1

Top-up restores collection

Objective: After lockout, crediting escrow allows collection again.
Preconditions: Agent A2 locked out (remaining $= 0$); cashier/admin can post MVP top-up.
Steps: 1) Confirm lockout. 2) Post top-up of 20,000 XAF on A2. 3) Agent refreshes escrow. 4) Collect 5,000 XAF with GPS on.
Expected: Remaining shows 20,000 then 15,000; collection succeeds.
Refs: FR-03 (MVP path), FR-04.

TC-ESC-05 P1

Temporary ceiling override with audit

Objective: Only authorised role can raise ceiling; action is audited.
Preconditions: A2 at lockout; user Admin has override right; user Agent does not.
Steps: 1) As Agent, attempt override API/UI — expect deny. 2) As Admin, set temporary ceiling +30,000 XAF with reason. 3) Open audit log.
Expected: Agent denied; Admin succeeds; audit entry with actor, old/new ceiling, timestamp.
Refs: FR-05, UC-05.

TC-ESC-06 P0

Boundary — collection equals remaining escrow

Objective: Exact ceiling consumption is allowed and leaves remaining $= 0$.
Preconditions: A1 remaining escrow synchronised to exactly 10,000 XAF; client C01 available.
Steps: 1) Start collection of 10,000 XAF. 2) Enter denomination lines summing to 10,000. 3) GPS on; confirm.
Expected: Transaction accepted; remaining escrow $= 0$; further collection blocked (as TC-ESC-02).
Refs: FR-04, BR-03.

TC-ESC-07 P0

Boundary — collection exceeds remaining by 1 XAF

Objective: Amount remaining$+1$ is rejected.
Preconditions: A1 remaining $= 10{,}000$ XAF.
Steps: Attempt collection of 10,001 XAF with valid denomination and GPS.
Expected: Rejected before commit; remaining still 10,000; no financial row.

Refs: FR-04, BR-03.

TC-ESC-08 P0

Boundary — remaining−1 still allowed
Objective: Amount one unit under remaining succeeds.
Preconditions: A1 remaining = 10,000 XAF.
Steps: Collect 9,999 XAF; valid denomination; GPS on.
Expected: Success; remaining = 1 XAF.
Refs: FR-04, BR-03.

TC-COL

TC-COL-01 P0

Happy path online collection
Objective: End-to-end online deposit with all mandatory fields.
Preconditions: A1 logged in; network on; remaining escrow sufficient; client C01; GPS enabled with valid fix.
Steps: 1) Identify C01. 2) Enter amount 2,000 XAF. 3) Enter denomination lines that sum to 2,000. 4) Confirm GPS indicator is valid. 5) Submit.
Expected: Success; server row with UTC timestamp and non-null lat/lon; denomination lines persisted; escrow reduced by 2,000.
Refs: FR-06, FR-08, FR-12, BR-02, BR-05.

TC-COL-02 P0

GPS off blocks collection
Objective: BR-05 is enforced on the device and cannot be skipped.
Preconditions: A1 logged in; location services turned *off* at OS level.
Steps: 1) Open collection. 2) Select client and amount with valid denomination. 3) Attempt confirm.
Expected: UI blocks submit with clear message; if API is forced, server rejects; no financial row created.
Refs: FR-12, BR-05.

TC-COL-03 P0

Invalid GPS fix rejected
Objective: Collection requires a usable fix (accuracy within policy).
Preconditions: Agent logged in; mock or device returns accuracy > configured threshold (e.g. $> 50\,\text{m}$) or null coordinates.
Steps: 1) Start collection. 2) Enter amount and valid denomination. 3) Attempt confirm while fix is invalid.
Expected: Submit blocked or API 422; no financial row until accuracy is acceptable.
Refs: FR-12, BR-05, Geolocation Spec.

TC-COL-04 P0

Denomination sum mismatch
Objective: Declared amount must equal the sum of denomination lines.
Preconditions: A1 logged in; GPS on; escrow sufficient; client C01.
Steps: 1) Identify C01. 2) Enter amount 10,000 XAF. 3) Enter denomination lines that sum to 9,500 XAF only. 4) Attempt confirm.
Expected: Validation error before commit; no financial row; escrow unchanged.
Refs: FR-08, BR-02.

TC-COL-05 P0

Client identification by member ID and phone
Objective: Agent can resolve the correct client before amount entry.
Preconditions: Clients C01 (known member ID) and C02 (known phone) in seed data.
Steps: 1) Search by member ID of C01 — select. 2) Clear. 3) Search by phone of C02 — select.
Expected: Correct client name/ID shown in context for each search; no cross-identity mix-up.
Refs: FR-06.

TC-COL-06 P0

Geotag immutable after save
Objective: Lat/lon recorded at collection cannot be altered by agent or API.
Preconditions: TC-COL-01 completed; note stored coordinates.
Steps: 1) Attempt UI edit of coordinates (if any control exists). 2) Attempt `PATCH`/update with different lat/lon on collection id.
Expected: UI has no edit; API returns 403/405/422; original geotag unchanged in DB.
Refs: FR-12.

TC-OFF

TC-OFF-01 P0

Offline collection queued encrypted
Objective: Deposits work without network and are stored securely on device.
Preconditions: A1 logged in; GPS on; remaining escrow sufficient; enable airplane mode.
Steps: 1) Confirm offline state. 2) Complete one valid collection. 3) Observe UI pending/sync indicator. 4) Optionally inspect that DB file is not cleartext (see TC-SEC-04).
Expected: Collection accepted locally; marked unsynced; encrypted at rest; visible as pending until sync.
Refs: FR-07, NFR-01.

TC-OFF-02 P0

Sync uploads once (idempotency)
Objective: Retries must not double-post the same device transaction.
Preconditions: One offline collection pending with known `device_tx_id`; network restorable.

Steps: 1) Restore network. 2) Trigger sync. 3) Trigger sync again immediately. 4) Query server by `device_tx_id`.
Expected: Exactly one server row; second sync is no-op or 200 without duplicate amount/escrow effect.
Refs: FR-07, NFR-08.

TC-OFF-03 P0

Offline burst of three collections

Objective: Multiple offline deposits sync completely once.
Preconditions: Agent online escrow sufficient; then airplane mode.
Steps: 1) Record three valid offline collections (different clients). 2) Restore network. 3) Sync once.
Expected: Three server rows; escrow reduced by sum of three; timestamps ordered.
Refs: FR-07.

TC-OFF-04 P0

Failed sync retains local queue

Objective: Server errors must not drop offline data.
Preconditions: At least one unsynced local collection; ability to mock API 500.
Steps: 1) Trigger sync against failing endpoint. 2) Confirm local pending remains. 3) Restore API; sync again.
Expected: First attempt fails without purge; second attempt posts once (idempotent).
Refs: FR-07, NFR-08.

TC-GEO

TC-GEO-01 P0

Background pulse stages points offline

Objective: Continuous trail is captured without network and without continuous GPS drain.
Preconditions: Field session started; background location permission granted; airplane mode on.
Steps: 1) Start session. 2) Remain offline for at least 15 minutes while moving or idle. 3) Inspect local `local_agent_tracking` (or equivalent).
Expected: Rows approximately every 5 minutes (or stationary recycle entries); timestamps in UTC; data encrypted at rest; no crash.
Refs: FR-10, FR-07, NFR-09, Geolocation Spec.

TC-GEO-02 P0

Batch location sync returns 202

Objective: Offline trail flushes asynchronously without blocking the agent.
Preconditions: $\geq$3 unsynced local tracking points; network available; valid agent JWT.
Steps: 1) Trigger batch location upload (automatic or manual). 2) Observe HTTP status. 3) Check server trail store / queue processing. 4) Check local flags purged or `is_synced=true`.
Expected: API returns 202 Accepted; points appear in PostgreSQL after consumer process-

ing; local pending cleared; duplicates avoided via `tracking_id`.
Refs: FR-10, Geolocation Spec.

TC-GEO-03 P1

Map shows last known during blackout
Objective: Manager UI must not invent movement while agent is offline.
Preconditions: Agent had at least one online ping; then offline $\geq$10 minutes with local pulses.
Steps: 1) Open back-office map during blackout. 2) Note marker position. 3) After sync, re-open map.
Expected: During blackout, marker at last online point only; after sync, FR-11 polyline includes offline milestones.
Refs: FR-11, Geolocation Spec.

TC-GEO-04 P1

Tracking stops outside service hours
Objective: No new trail points after configured end of service window.
Preconditions: Branch window ends at 18:00 WAT; agent session was active.
Steps: 1) Set device clock (or config) past end of window. 2) Wait for two pulse intervals. 3) Inspect local tracking table.
Expected: No new rows after window end; optional banner that tracking is off-duty.
Refs: NFR-10.

TC-OFJ

TC-OFJ-01 P0

Balanced end-of-day close
Objective: When physical cash equals system total, OFJ can close cleanly.
Preconditions: Day's collections synced; system total S displayed on OFJ screen; cashier role.
Steps: 1) Open OFJ for branch. 2) Enter physical count $= S$. 3) Confirm close.
Expected: $\Delta = 0$; session status closed; export job enqueued toward CBS mock (FR-18); audit trail written.
Refs: FR-16, FR-18, BR-01.

TC-OFJ-02 P0

Negative variance blocks close
Objective: Physical count below system total cannot be ignored.
Preconditions: System total S known; agent/cashier role on OFJ screen.
Steps: 1) Enter physical count $S - 1{,}000$. 2) Attempt close.
Expected: Close refused or investigation ticket forced; agent debt recorded; audit entry; no silent success.
Refs: FR-16, FR-17, BR-01.

TC-OFJ-03 P0

Positive variance handled per policy

Objective: Count above system total follows documented business rule.
Preconditions: System total S.
Steps: Enter physical count $S + 500$; attempt close.
Expected: Behaviour matches FR text (flag for investigation and/or allowed close with reason); always audited.
Refs: FR-16, FR-17.

TC-OFJ-04 P0

CBS export stub after successful OFJ

Objective: Closed day triggers export toward middleware/mock.
Preconditions: TC-OFJ-01 success path available.
Steps: 1) Complete balanced OFJ. 2) Observe export job / mock endpoint / file. 3) Re-trigger export.
Expected: Payload received once per logical day export; second call idempotent (no duplicate ledger effect).
Refs: FR-18.

TC-CLI

TC-CLI-01 P0

Client views own profile only

Objective: Authenticated member sees only their profile.
Preconditions: Client user for C01; another member C02 exists.
Steps: 1) Login as C01. 2) Open profile. 3) Attempt access to C02 profile URL/id if exposed.
Expected: C01 data shown; C02 access denied or not reachable.
Refs: FR-20.

TC-CLI-02 P0

Client views balance after sync

Objective: Balance reflects posted collections.
Preconditions: Known collection for C01 synced to server.
Steps: 1) Login as C01. 2) Open balance view. 3) Compare with back-office/server total for C01.
Expected: Amounts match within defined rounding (integer XAF).
Refs: FR-21.

TC-CLI-03 P1

Contribution history list

Objective: Client can list past contributions.
Preconditions: C01 has $\geq$2 synced collections; C03 has zero.
Steps: 1) Login C01 — open history. 2) Login C03 — open history.
Expected: C01 sees ordered list (newest first); C03 sees empty state message.
Refs: FR-22.

TC-SEC

TC-SEC-01 P0

Protected API rejects missing JWT
Objective: Unauthenticated callers cannot collect or read private data.
Preconditions: Gateway/Core running; collection endpoint known from Swagger.
Steps: 1) `POST /api/v1/collections` without `Authorization` header. 2) Repeat with garbage token.
Expected: HTTP 401; no row created.
Refs: FR-01 (session), System Design security.

TC-SEC-02 P0

Broken object-level authorization — cross-branch deny
Objective: Agent of branch A cannot read or modify branch B resources.
Preconditions: Agent A1 bound to `BR-YDE-01`; resource IDs exist for branch `BR-DLA-02` (seed or fixture).
Steps: 1) Login as A1; capture JWT. 2) `GET` escrow or collections for a `BR-DLA-02` agent/client id. 3) Attempt `POST` collection attributed to the other branch.
Expected: 403 Forbidden or empty/not-found without data leak; no write side-effect.
Refs: STRIDE Elevation / BOLA; System Design.

TC-SEC-03 P1

Gateway rate limiting
Objective: Burst traffic receives 429 before Core is overwhelmed.
Preconditions: Rate limit configured (e.g. N req/min on login or public route).
Steps: 1) Script or repeat requests above threshold from one IP. 2) Observe status codes.
Expected: Excess calls return 429; legitimate spaced calls still succeed after window.
Refs: System Design — DoS control.

TC-SEC-04 P0

Offline database not readable as cleartext
Objective: Local collection/tracking store is encrypted at rest.
Preconditions: At least one offline collection stored; device file access (debug build).
Steps: 1) Locate app DB file. 2) Open without key as raw bytes/text. 3) Search for known client name or amount string.
Expected: No plaintext match; content appears ciphertext/unreadable.
Refs: NFR-01, STRIDE Disclosure/Tampering.

TC-UX — Usability smoke

TC-UX-01 P0

Agent completes one deposit under five minutes
Objective: Primary field path is efficient for non-expert agents.
Preconditions: A1 logged in; GPS on; client C01 known; escrow sufficient.
Steps: 1) Start timer. 2) Identify C01. 3) Enter amount 2,000 XAF and matching denomination. 4) Confirm. 5) Stop timer on success feedback.
Expected: End-to-end $\leq$5 minutes on first assisted attempt; no dead-end screens; success feedback visible.
Refs: NFR-06 (inclusive ergonomics); Test Plan usability smoke type.

Traceability (MVP and referenced FRs)

FR/BR	Topic	Cases	P
FR-01	SSO / IMEI	TC-AUTH-01..03, TC-SEC-01	P0
FR-02	Enrollment	TC-AUTH-04	P0
FR-03	Escrow top-up (MVP path)	TC-ESC-04	P1
FR-04 / BR-03	Escrow lockout + boundaries	TC-ESC-01..03, TC-ESC-06..08	P0
FR-05	Ceiling override	TC-ESC-05	P1
FR-06	Client identification	TC-COL-05	P0
FR-07	Offline mode	TC-OFF-01..04, TC-ESC-03	P0
FR-08 / BR-02	Denomination	TC-COL-01, TC-COL-04	P0
FR-10	Trail geolocation	TC-GEO-01..02	P0
FR-11	Route / map	TC-GEO-03	P1
FR-12 / BR-05	GPS gate / geotag	TC-COL-02,03,06	P0
FR-16 / BR-01	OFJ reconcile	TC-OFJ-01..03	P0
FR-17	Variance debt	TC-OFJ-02	P0
FR-18	CBS export stub	TC-OFJ-04	P0
FR-20	Client profile	TC-CLI-01	P0
FR-21	Client balance	TC-CLI-02	P0
FR-22	History	TC-CLI-03	P1
NFR-01	AES-256 offline	TC-OFF-01, TC-SEC-04	P0
NFR-06	Ergonomics	TC-UX-01	P0
NFR-08	Sync behaviour	TC-OFF-02..04	P0
NFR-09	Battery pulse	TC-GEO-01	P0
NFR-10	Off-duty privacy	TC-GEO-04	P1
STRIDE	BOLA / DoS	TC-SEC-02, TC-SEC-03	P0/P1

Table 1: Requirement-to-test traceability. MVP P0 rows support the exit criterion "every MVP FR mapped". FR-09, FR-13–15, FR-19 remain V1.1/V2 (no P0 case required for freeze).

5. Test Procedure Specification

Identifier

`TPS-MICROFI-V1.0`

Purpose

Define the **sequence** in which related test cases are executed so setup is shared and dependencies are respected.

Procedure PROC-SMOKE — Build smoke (30 min)

1. Transmit build (Section 6).
2. Start Docker stack; open Swagger; `GET /health`.
3. TC-AUTH-01 on seed agent A1.
4. TC-COL-01 one online collection.
5. Log results in Test Log.

Procedure PROC-MONEY — Money path (P0)

Order: TC-AUTH-01 → TC-ESC-01 → TC-ESC-08 → TC-ESC-06 → TC-ESC-07 → TC-COL-01 → TC-COL-04 → TC-ESC-02 → TC-COL-02.
Special: Reset escrow between boundary ESC cases; use remaining $= 10{,}000$ XAF fixture for TC-ESC-06..08.

Procedure PROC-OFFLINE — Offline & sync (P0)

Order: TC-AUTH-01 → TC-OFF-01 → TC-OFF-03 → TC-OFF-02 → TC-OFF-04 → TC-ESC-03.
Special: Airplane mode; do not clear app data mid-procedure.

Procedure PROC-OFJ — End of day (P0)

Order: Ensure day's collections synced → TC-OFJ-01 → TC-OFJ-02 → TC-OFJ-04.

Procedure PROC-GEO — Trail (P0/P1)

Order: Start field session → TC-GEO-01 (wait $\geq$15 min offline) → TC-GEO-02.

Procedure PROC-ACCEPT — Acceptance demo

Duration $\approx$15–20 min on physical Android.

1. Login A1; reject wrong PIN once (AUTH-01/02).
2. Online collection with denomination + GPS (COL-01).
3. GPS off → blocked (COL-02).
4. Escrow lockout (ESC-02).
5. Offline two collections; sync once; no duplicates (OFF).
6. OFJ balanced close; optional negative variance (OFJ).
7. Client profile + balance (CLI).

Pass: All steps without Critical defect; GPS gate and lockout not bypassable.

6. Test Item Transmittal Report

> Use this form whenever development hands a build to testing (or self-test for academic solo work).

Field	Value
Report ID	TITR-MICROFI-____
Date / time	
Transmitter (dev)	
Receiver (test)	
Build / commit / tag	
Artifacts	APK / Docker images / OpenAPI revision
Environment	Local / Staging
Features included	(e.g. FR-07 offline sync)
Known limitations	
Smoke status before handoff	Pass / Fail
Location of binaries	
Signature transmitter	
Signature receiver	

Rule: No acceptance procedure starts without a completed transmittal for the build under test.

7. Test Log

Identifier

`TL-MICROFI-`[date]

Purpose

Chronological record of who ran which procedure/case, when, on which build, and the result.

Date/Time	Tester	Build	Proc/TC	Result	Notes
				Pass / Fail	
				Pass / Fail	
				Pass / Fail	
				Pass / Fail	

Table 2: Test Log template — duplicate rows as needed.

Status values: Pass, Fail, Blocked, Not Run, Skipped.

8. Test Incident Report

File one report per deviation from expected results. Link to Test Log entry and affected TC/FR.

Field	Value
Incident ID	INC-MICROFI-____
Date detected	
Tester	
Build / commit	
Procedure / TC	
Related FR/BR	
Severity	Critical / Major / Minor / Trivial
Summary	
Steps to reproduce	
Expected result	
Actual result	
Environment	Device, OS, network mode
Attachments	Logs, screenshots
Status	New / Confirmed / Fixed / Verified / Closed
Assignee	

Severity guide

Critical: wrong amount, double post, lockout bypass, collection without GPS.
Major: sync loss, wrong OFJ delta, crash on main path.
Minor / Trivial: UI wording, cosmetic.

9. Test Summary Report

Identifier

`TSR-MICROFI-V1.0` — completed at MVP freeze / acceptance.

> **Pre-execution status.** This section is a *template* until a formal test cycle is run. Pass/fail counts, dates, and the evaluation narrative are filled from the Test Log and Incident Reports after execution. Exit criteria in the Test Plan become *evidence* only once this report is completed with real results — not while fields remain blank.

Summary

Field	Entry
Project / version	MICROFI Collect V1.0 MVP
Test period	From ____ to ____
Build(s) tested	
Procedures executed	PROC-SMOKE / MONEY / OFFLINE / OFJ / GEO / ACCEPT
P0 cases planned	
P0 passed / failed / blocked	
Incidents opened / closed	
Critical open at freeze	0 required for acceptance
Environment	

Evaluation

- Features that met pass criteria: ____
- Features deferred: ____
- Residual risks: GPS device dependency; CBS mock only; battery NFR qualitative.

Summary of results

> *(Narrative: overall quality judgment, readiness for academic demo / pilot.)*

Recommendations

1. Accept / conditional accept / reject MVP freeze.
2. Priority fixes before pilot: ___________________________________
3. Next test cycle focus (V1.1): FR-05, FR-09, FR-13, FR-19, FR-22.

Approvals

Author: ____________________	Supervisor: ____________________	Date: ______________

10. Automation Notes

- **Backend:** JUnit / WebTestClient for escrow, denomination, GPS filter, idempotent sync.
- **API: Swagger UI / OpenAPI (springdoc)** for interactive contract testing; include negative cases.
- **Mobile:** Queue persistence tests; GPS on physical device.
- **CI:** Unit + API smoke on merge; full device suite before acceptance procedure.

MICROFI Collect — PK Fokam Research Center / ENSPY 4GI

Test Documentation v1.3 — Full IEEE 829–style pack

Design system aligned with Business Analysis v3.1